ProMarket Supporting Methodology and Results

Supporting verification pack for "Bank Concentration Undermined The Fed's Monetary Policy During the Last Tightening Cycle"

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Purpose: document the data construction, calculations and robustness checks behind the ProMarket article. This is supporting material for editorial verification, not a separate working paper.

Verification summary

Item	Result
Main descriptive panel	91,105 usable bank-quarter observations, 2021Q3-2026Q2
HHI coverage	91,397 of 92,586 analysis rows matched to SOD concentration; 98.72%
Endpoint beta sample	4,168 banks observed at 2021Q4, 2024Q2 and 2026Q2
Balanced FE sample	4,123 banks observed in every quarter from 2021Q4 to 2026Q2
Extreme-quartile FE sample	2,064 banks in the bottom/top 2021Q4 HHI quartiles
Tightening/hold result	Concentrated group: 7.03bp less deposit-rate response per 100bp EFFR movement
Easing result	Concentrated group: 8.02bp less deposit-rate response per 100bp EFFR movement

1. Data sources and sample construction

The analysis combines three public U.S. data sources: FFIEC Call Reports for bank financial statements, the FDIC Summary of Deposits (SOD) for branch-level deposit geography, and the Federal Reserve Bank of New York effective federal funds rate (EFFR).

Source	Frequency	Role	Official source
FFIEC Call Reports	Quarterly	Deposit interest expense, interest-bearing domestic deposits, total assets	https://cdr.ffiec.gov/public/pws/downloadbulkdata.aspx
FDIC Summary of Deposits	Annual, June 30	Branch deposits used to construct county HHI and bank deposit-weighted HHI	https://banks.data.fdic.gov/bankfind-suite/SOD
New York Fed EFFR	Daily	Quarterly average policy-rate benchmark	https://www.newyorkfed.org/markets/reference-rates/effr

Sample flow

Check	Result
Rows entering analysis	92,586
Rows with SOD/HHI match	91,397
HHI merge rate	98.72%
Final usable bank-quarter rows	91,105
Identifier agreement check	99.9934%

The descriptive sample runs from 2021Q3 through 2026Q2. For cumulative betas, the common endpoint sample requires a bank to be observed at 2021Q4, 2024Q2 and 2026Q2. For fixed-effects regressions, the balanced sample requires presence in all 19 quarters from 2021Q4 through 2026Q2.

2. Variable construction

Effective deposit funding rate

Numerator. Domestic deposit interest expense equals RIAD4508 (interest-bearing transaction accounts) + RIAD0093 (savings/MMDAs) + RIADHK03 (time deposits <= $250,000) + RIADHK04 (time deposits > $250,000). RIAD4172, interest on deposits in foreign offices, is excluded. Call Report income items are year-to-date and are differenced to obtain quarterly flows.

Denominator and annualization. The main panel uses the average of adjacent quarter-end interest-bearing domestic deposit balances. The annualized effective rate is quarterly domestic deposit interest expense divided by average interest-bearing domestic deposits, multiplied by 400. This is an effective realized funding cost across the bank's existing stock of interest-bearing domestic deposits, not the advertised APY on a particular product.

RC-K validation: in a June 2026 cross-check covering 4,227 banks, the median ratio of the shortcut denominator to the reported quarterly-average denominator was about 1.0006, and the median rate difference was about -0.12 basis points.

Deposit-market concentration

For each SOD year, branch deposits (DEPSUMBR) are aggregated by bank within county. County HHI is the sum of squared bank deposit shares. A bank-level HHI is then constructed as the deposit-weighted average of the county HHIs across the bank's branch footprint. HHI is stored on a 0-1 scale; 0.10 corresponds to 1,000 conventional HHI points. The latest SOD vintage available without look-ahead is applied to each quarter.

For endpoint and regression comparisons, competitive and concentrated groups are fixed using the bottom and top quartiles of 2021Q4 bank HHI, respectively. Fixing classification at the start of the cycle avoids mechanically reclassifying banks as the policy cycle unfolds.

3. Descriptive evidence across the cycle

The competition gap is defined as the median effective deposit funding rate of bottom-HHI-quartile banks minus the median rate of top-HHI-quartile banks. It is close to zero before tightening, widens sharply as EFFR rises, and narrows during easing while remaining positive through 2026Q2.

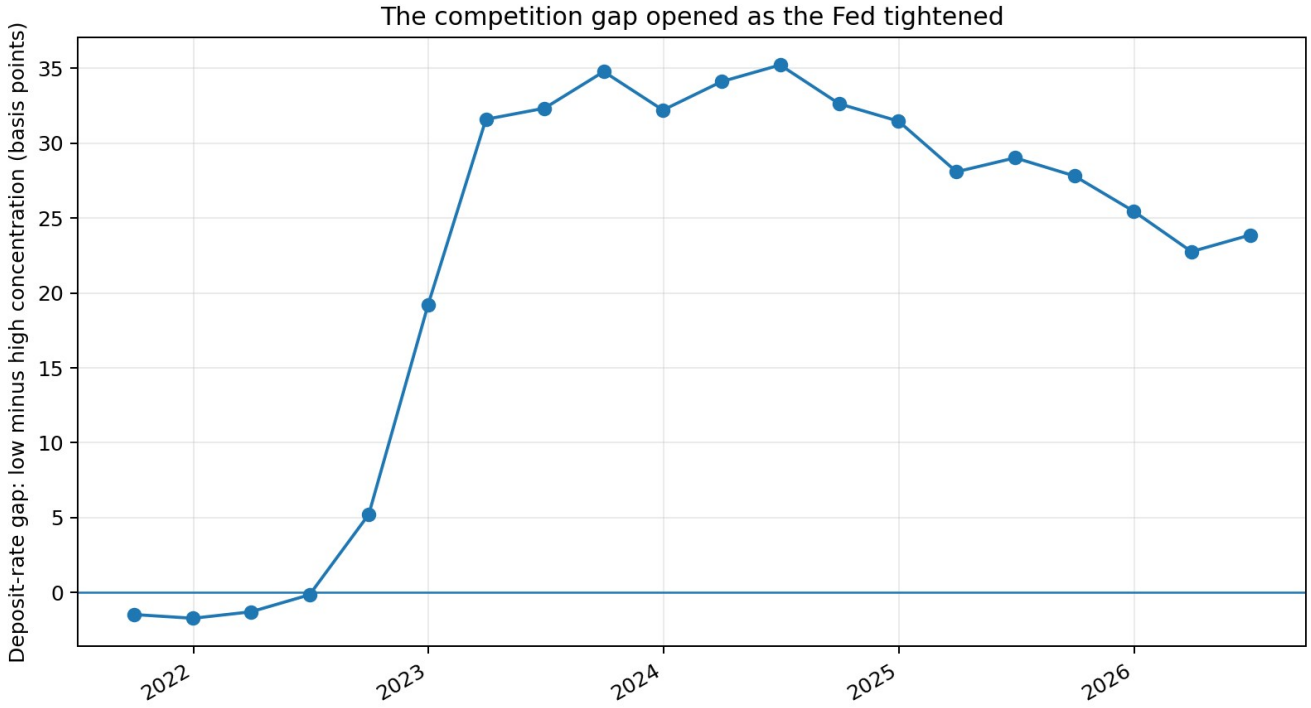


Figure 1. Competition gap by quarter. Positive values indicate higher effective deposit funding rates among banks operating in less concentrated markets.

Quarter	EFFR %	Low-HHI median %	High-HHI median %	Gap bp	N
2021-09-30	0.08	0.37	0.39	-1.47	4,893
2021-12-31	0.08	0.33	0.35	-1.72	4,807
2022-12-31	3.65	0.89	0.70	19.20	4,678
2023-06-30	4.99	1.83	1.51	32.34	4,621
2023-12-31	5.33	2.48	2.16	32.22	4,570
2024-06-30	5.33	2.81	2.46	35.23	4,522
2024-12-31	4.66	2.84	2.52	31.48	4,467
2025-12-31	3.90	2.59	2.33	25.47	4,317
2026-06-30	3.63	2.44	2.20	23.87	4,215

The exact quarterly series is provided in the accompanying workbook. The gap reached 35.23bp in 2024Q2 and was 23.87bp in 2026Q2. At the same time, median effective funding rates rose from approximately 0.33% to 2.81% for the lower-concentration group and from approximately 0.35% to 2.46% for the higher-concentration group between late 2021 and mid-2024.

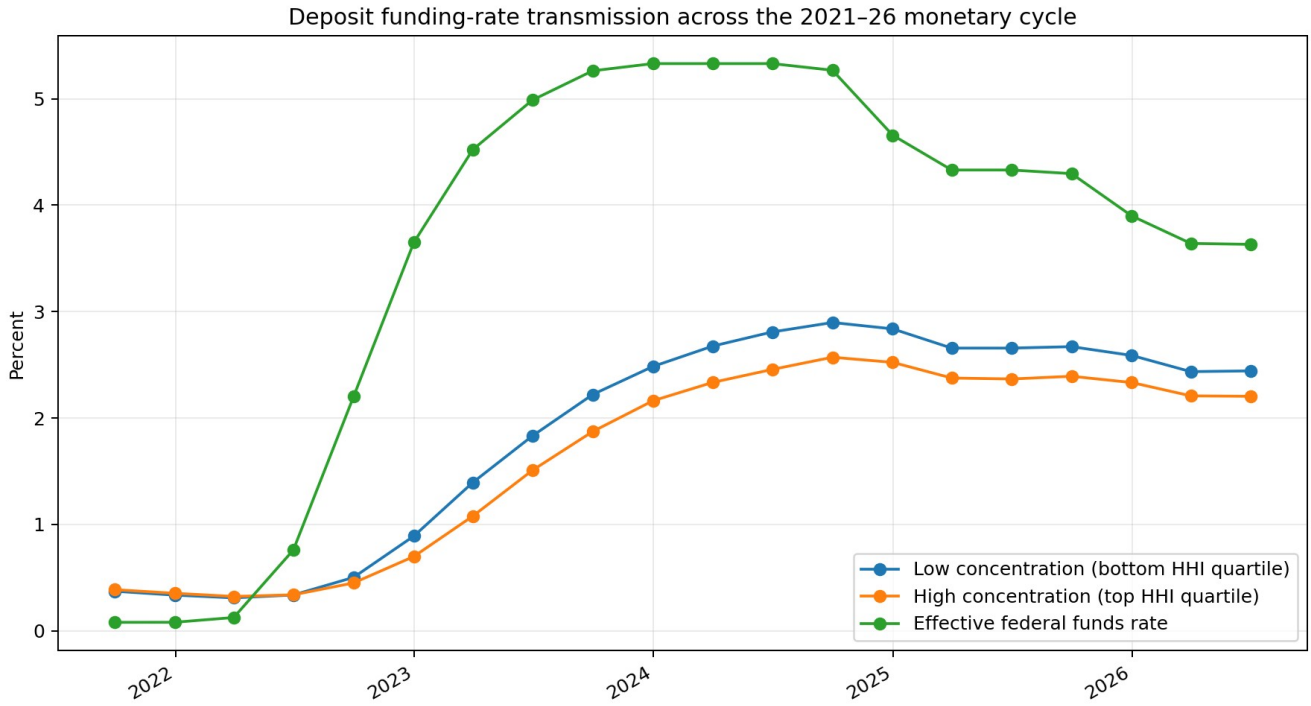


Figure 2. Median effective deposit funding rates by concentration group and quarterly average EFFR.

4. Cumulative deposit betas

A cumulative deposit beta is the change in a group's median effective deposit funding rate divided by the change in EFFR over the same episode. The calculation uses a common sample of 4,168 banks observed at all three relevant endpoints and fixes HHI classification using late-2021 exposure.

Group	Episode	Start dep. rate %	End dep. rate %	EFFR change pp	Beta
Competitive	Tightening/hold	0.338	2.754	5.250	46.0%
Competitive	Easing	2.754	2.427	-1.699	19.3%
Concentrated	Tightening/hold	0.352	2.454	5.250	40.0%
Concentrated	Easing	2.454	2.198	-1.699	15.1%

The tightening/hold beta is 46.03% for lower-concentration banks and 40.04% for higher-concentration banks. During easing, the corresponding betas are 19.28% and 15.09%. The easing betas are lower in level, but the two episodes differ materially in length and size; the analysis therefore uses them primarily to compare the relative sensitivity of the two concentration groups within each episode.

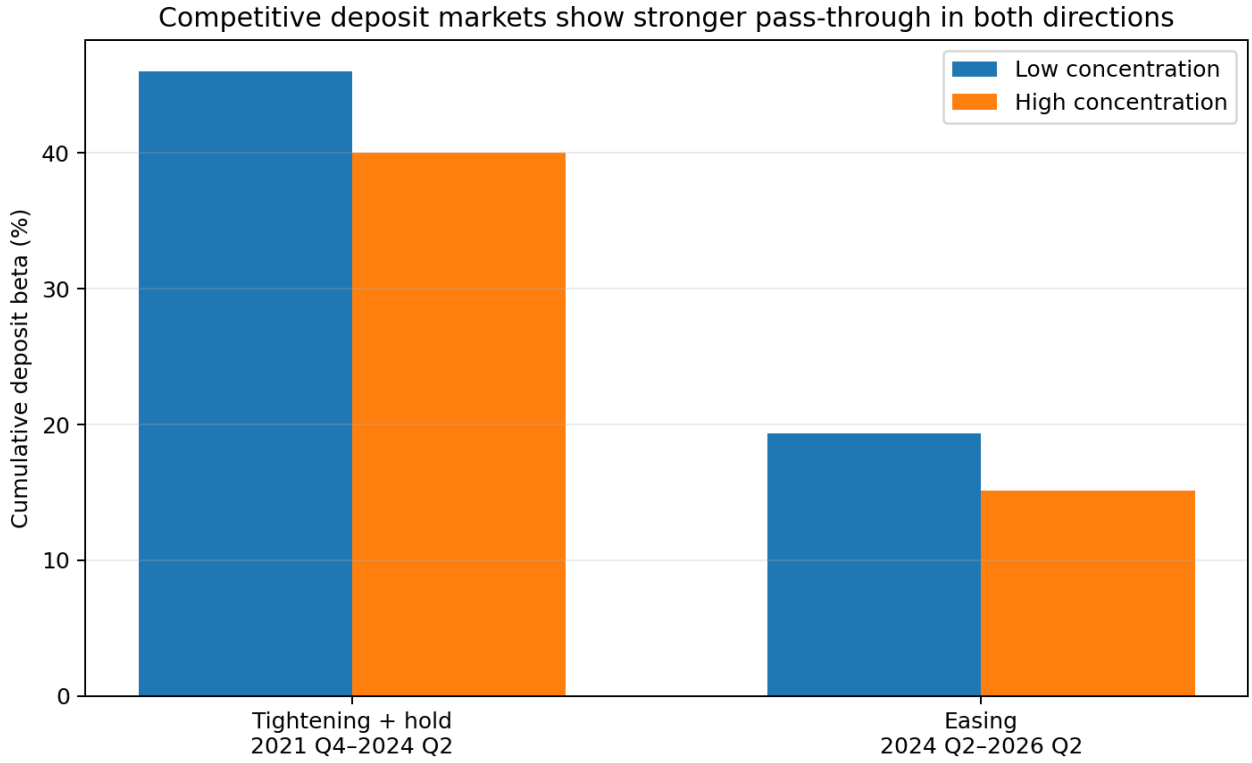


Figure 3. Cumulative deposit betas by concentration group and policy episode.

5. Fixed-effects robustness

The main robustness specification estimates the differential sensitivity of deposit funding rates to EFFR while absorbing persistent differences across banks and common quarter shocks:

DepositRate_it = alpha_i + gamma_t + beta (EFFR_t x Concentration_i,2021Q4) + delta ln(Assets_it) + error_it

Bank and quarter fixed effects are included, total assets enter in logs, and standard errors are clustered by bank. The extreme-quartile model uses a fixed indicator for the top versus bottom 2021Q4 HHI quartile. Because the indicator is time-invariant it is absorbed by bank fixed effects, while EFFR itself is absorbed by quarter fixed effects; the interaction identifies the differential pass-through.

Model	Episode	Banks	Obs.	Raw beta	SE	Less response bp	95% CI bp	p
Extreme quartiles	Tightening/ hold	2,064	22,704	-0.0703	0.0066	7.03	[5.74, 8.33]	2.17e-26
Extreme quartiles	Easing	2,064	18,576	-0.0802	0.0121	8.02	[5.64, 10.39]	3.82e-11
Continuous HHI	Tightening/ hold	4,123	45,353	-0.1615	0.0184	1.62	[1.25, 1.98]	1.62e-18

Continuous HHI	Easing	4,123	37,107	-0.1964	0.0328	1.96	[1.32, 2.61]	2.1e-09

Interpretation: in the extreme-quartile model, higher-concentration banks show 7.03bp less deposit-rate response per 100bp EFFR movement during tightening/hold and 8.02bp less during easing. In the continuous model, an additional 1,000 conventional HHI points is associated with 1.62bp less response during tightening/hold and 1.96bp less during easing.

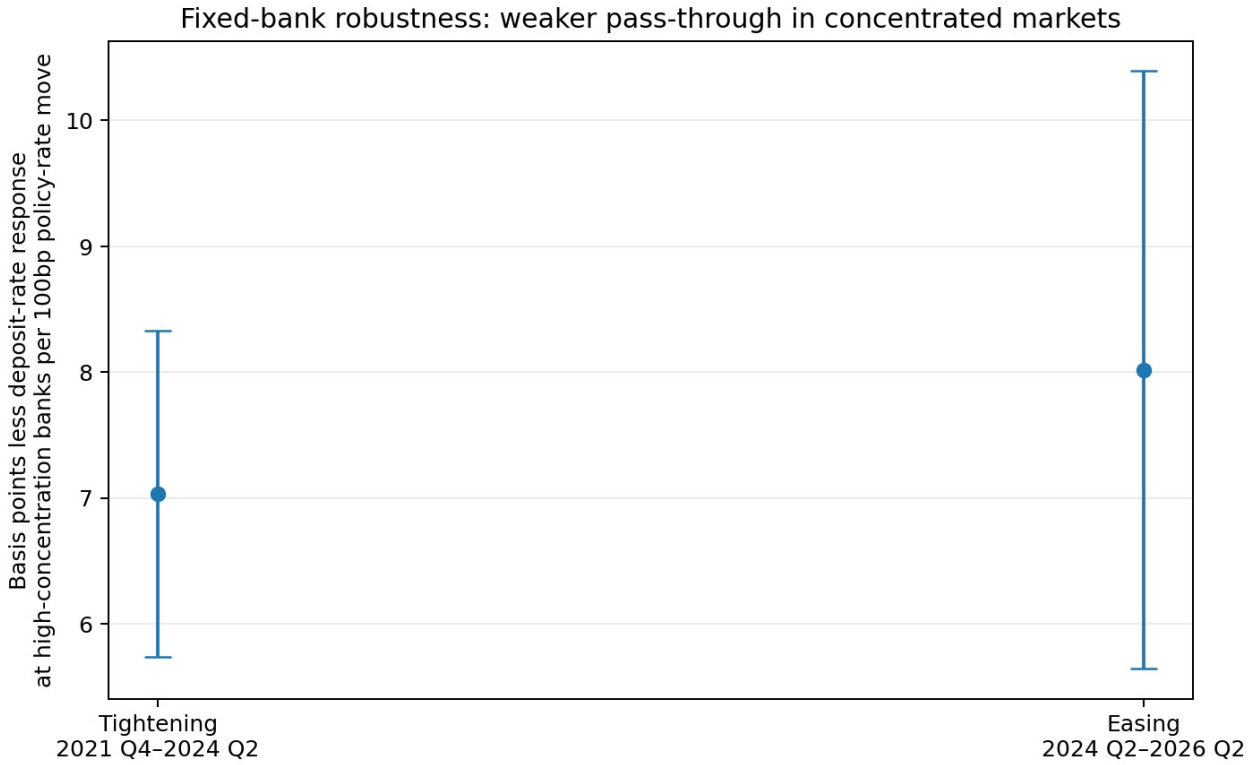


Figure 4. Extreme-quartile fixed-effects estimates and 95% confidence intervals.

6. Sensitivity checks

The same extreme-quartile fixed-effects design was re-estimated after restricting the 2021Q4 baseline sample. Quartile cutoffs are recalculated within each restricted sample. The sign and broad magnitude are stable when very large banks are excluded, when very high-HHI observations are excluded, and in a sample restricted to banks with at least $300 million in assets. In the smaller >=$1 billion sample, the tightening estimate remains negative and statistically precise, while the easing estimate remains negative but is imprecise.

Check	Episode	Banks	Less response bp	SE bp	95% CI	p
Baseline extreme quartiles	Tightening/hold	2,064	7.03	0.66	[5.74, 8.33]	2.17e-26
Baseline extreme quartiles	Easing	2,064	8.02	1.21	[5.64, 10.39]	3.82e-11
Exclude banks > $100bn assets	Tightening/hold	2,051	7.02	0.66	[5.73, 8.31]	1.79e-26
Exclude banks > $100bn assets	Easing	2,051	8.09	1.21	[5.71, 10.46]	2.59e-11
Exclude baseline HHI > 0.50	Tightening/hold	1,975	6.81	0.68	[5.49, 8.14]	6.25e-24
Exclude baseline HHI > 0.50	Easing	1,975	7.41	1.25	[4.97, 9.85]	2.68e-09
Banks >= $300m assets	Tightening/hold	1,088	8.46	0.91	[6.68, 10.24]	1.32e-20
Banks >= $300m assets	Easing	1,088	9.24	1.56	[6.19, 12.30]	3.06e-09
Banks >= $1bn assets	Tightening/hold	410	6.16	1.60	[3.03, 9.28]	0.000114
Banks >= $1bn assets	Easing	410	3.69	2.68	[-1.56, 8.95]	0.168

7. Validation and interpretation boundaries

HHI coverage: 98.72% of analysis rows successfully match to an SOD-based concentration measure.

Identifier consistency: Certificate agreement is 99.9934% in the panel construction check.

Funding-rate denominator: RC-K validation indicates the adjacent-quarter-end denominator is very close to the reported quarterly average.

Measure interpretation: The effective funding rate is an average realized cost across interest-bearing domestic deposits and need not equal advertised product APYs.

Causality: The analysis establishes a robust association between market structure and repricing, not proof that concentration alone causes the difference.

HHI interpretation: HHI measures concentration, not market power itself.

Geography: County-level concentration is an imperfect proxy as online banks compete across local boundaries.

8. Reproducibility map

The accompanying ZIP contains the processed bank-quarter panel, exact quarterly summary, analysis summary, endpoint-beta table, fixed-effects regression table, sensitivity table, four article/robustness figures, a calculation workbook, and a Python reproduction script. The script reproduces the endpoint-beta and fixed-effects results directly from the saved processed panel and quarterly EFFR series.

File	Purpose
Data/promarket_bank_quarter_panel_mvp.csv	Processed 91,105-row analysis panel
Data/promarket_quarter_summary.csv	Exact quarter-by-quarter medians, EFFR, HHI cutoffs and gap
Data/promarket_analysis_summary.csv	Sample and headline summary metrics
Data/endpoint_betas.csv	Endpoint rates, EFFR changes and cumulative betas
Data/regression_results.csv	Main extreme-quartile and continuous-HHI FE estimates
Data/sensitivity_checks.csv	Restricted-sample FE robustness checks
ProMarket_Supporting_Calculations.xlsx	Human-readable calculation and number-mapping workbook
Code/reproduce_support_results.py	Reproduces endpoint betas and FE outputs from included

	processed data
Figures/*.png	Three article figures plus fixed-effects robustness figure

Source note. All empirical results are calculations using FFIEC Call Reports, FDIC Summary of Deposits and Federal Reserve Bank of New York EFFR data. Raw public downloads are not duplicated in this compact support pack; the official source locations are documented above, while the exact processed panel used for the article is included.